
MultiSync LCD1880SX

User's Manual

Bedienerhandbuch

Manual del usuario

Manuel Utilisateur

Manuale utente

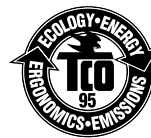
NEC



TCO'95

MultiSync LCD1880SX Black Model

Congratulations! You have just purchased a TCO'95 approved and labeled product! Your choice has provided you with a product developed for professional use. Your purchase has also contributed to reducing the burden on the environment and also, to the further development of environmentally adapted electronics products.



Why do we have environmentally labelled computers?

In many countries, environmental labelling has become an established method for encouraging the adaptation of goods and services to the environment. The main problem, as far as computers and other electronics equipment are concerned, is that environmentally harmful substances are used both in the products and during the manufacturing. Since it has not been possible for the majority of electronics equipment to be recycled in a satisfactory way, most of these potentially damaging substances sooner or later enter Nature.

There are also other characteristics of a computer, such as energy consumption levels, that are important from the viewpoints of both the work (Internal) and natural (external) environments. Since all methods of conventional electricity generation have a negative effect on the environment (acidic and climate-influencing emissions, radioactive waste, etc.), it is vital to conserve energy. Electronics equipment in offices consume an enormous amount of energy since they are often left running continuously.

What does labelling involve?

This product meets the requirements for the TCO'95 scheme which provides for international and environmental labelling of personal computers. The labelling scheme was developed as a joint effort by the TCO (The Swedish Confederation of Professional Employees), Naturskyddsforeningen (The Swedish Society for Nature Conservation) and NUTEK (The National Board for Industrial and Technical Development in Sweden).

The requirements cover a wide range of issues: environment, ergonomics, usability, emission of electrical and magnetic fields, energy consumption and electrical and fire safety.

The environmental demands concern restrictions on the presence and use of heavy metals, brominated and chlorinated flame retardants, CFCs (freons) and chlorinated solvents, among other things. The product must be prepared for recycling and the manufacturer is obliged to have an environmental plan which must be adhered to in each country where the company implements its operational policy. The energy requirements include a demand that the computer and/or display, after a certain period of inactivity, shall reduce its power consumption to a lower level in one or more stages. The length of time to reactivate the computer shall be reasonable for the user.

Labelled products must meet strict environmental demands, for example, in respect of the reduction of electric and magnetic fields, physical and visual ergonomics and good usability.

TCO'95 is a co-operative project between TCO (The Swedish Confederation of Professional Employees), Naturskyddsforeningen (The Swedish Society for Nature Conservation) and NUTEK (The National Board for Industrial and Technical Development in Sweden).

Environmental Requirements

Brominated flame retardants

Brominated flame retardants are present in printed circuit boards, cables, wires, casings and housings. In turn, they delay the spread of fire. Up to thirty percent of the plastic in a computer casing can consist of flame retardant substances. These are related to another group of environmental toxins, PCBs, which are suspected to give rise to similar harm, including reproductive damage in fish-eating birds and mammals, due to the bio-accumulative* processes. Flame retardants have been found in human blood and researchers fear that disturbances in foetus development may occur.

TCO'95 demand requires that plastic components weighing more than 25 grams must not contain organically bound chlorine and bromine.

Lead**

Lead can be found in picture tubes, display screens, solders and capacitors. Lead damages the nervous system and in higher doses, causes lead poisoning.

TCO'95 requirement permits the inclusion of lead since no replacement has yet been developed.

Cadmium**

Cadmium is present in rechargeable batteries and in the colourgenerating layers of certain computer displays. Cadmium damages the nervous system and is toxic in high doses.

TCO'95 requirement states that batteries may not contain more than 25 ppm (parts per million) of cadmium. The colourgenerating layers of display screens must not contain any cadmium.

Mercury**

Mercury is sometimes found in batteries, relays, switches, and back-light systems, Mercury damages the nervous system and is toxic in high doses.

TCO'95 requirement states that batteries may not contain more than 25 ppm (parts per million) of mercury. It also demands that no mercury is present in any of the electrical or electronics components concerned with the display unit, except the



TCO'95 –continued

back-light system.

CFCs (freons)

CFCs (freons) are sometimes used for washing printed circuit boards and in the manufacturing of expanded foam for packaging. CFCs break down ozone and thereby damage the ozone layer in the stratosphere, causing increased reception on Earth of ultraviolet light with consequent increased risks of skin cancer (malignant melanoma).

The relevant TCO'95 requirement; Neither CFCs nor HCFCs may be used during the manufacturing of the product or its packaging.

*Bio-accumulative is defined as substances which accumulate within living organisms.

**Lead, Cadmium and Mercury are heavy metals which are Bio-accumulative.

To obtain complete information on the environmental criteria document, order from:

TCO Development Unit
SE-114 94 Stockholm
SWEDEN
FAX Number: +46 8 782 92 07
E-mail (Internet): development@tco.se

You may also obtain current information on TCO'95 approved and labelled products by visiting their website at:
<http://www.tco-info.com/>

TCO'99

MultiSync LCD1880SX White Model

Congratulations! You have just purchased a TCO'99 approved and labeled product! Your choice has provided you with a product developed for professional use. Your purchase has also contributed to reducing the burden on the environment and also to the further development of environmentally adapted electronics products.



Why do we have environmentally labelled computers?

In many countries, environmental labelling has become an established method for encouraging the adaptation of goods and services to the environment. The main problem, as far as computers and other electronics equipment are concerned, is that environmentally harmful substances are used both in the products and during the manufacturing. Since it has not been possible for the majority of electronics equipment to be recycled in a satisfactory way, most of these potentially damaging substances sooner or later enter Nature.

There are also other characteristics of a computer, such as energy consumption levels, that are important from the viewpoints of both the work (Internal) and natural (external) environments. Since all methods of conventional electricity generation have a negative effect on the environment (acidic and climate-influencing emissions, radioactive waste, etc.), it is vital to conserve energy. Electronics equipment in offices consume an enormous amount of energy since they are often left running continuously.

What does labelling involve?

This product meets the requirements for the TCO'99 scheme which provides for international and environmental labelling of personal computers. The labelling scheme was developed as a joint effort by the TCO (The Swedish Confederation of Professional Employees), Svenska Naturskyddsforeningen (The Swedish Society for Nature Conservation) and Statens Energimyndighet (The Swedish National Energy Administration).

The requirements cover a wide range of issues: environment, ergonomics, usability, emission of electrical and magnetic fields, energy consumption and electrical and fire safety.

The environmental demands concern restrictions on the presence and use of heavy metals, brominated and chlorinated flame retardants, CFCs (freons) and chlorinated solvents, among other things. The product must be prepared for recycling and the manufacturer is obliged to have an environmental plan which must be adhered to in each country where the company implements its operational policy. The energy requirements include a demand that the computer and/or display, after a certain period of inactivity, shall reduce its power consumption to a lower level in one or more stages. The length of time to reactivate the computer shall be reasonable for the user.

Labelled products must meet strict environmental demands, for example, in respect of the reduction of electric and magnetic fields, physical and visual ergonomics and good usability.



TCO'99 –continued

Environmental Requirements

Flame retardants

Flame retardants are present in printed circuit boards, cables, wires, casings and housings. In turn, they delay the spread of fire. Up to thirty percent of the plastic in a computer casing can consist of flame retardant substances. Most flame retardants contain bromine or chloride and these are related to another group of environmental toxins, PCBs, which are suspected to give rise to severe health effects, including reproductive damage in fish-eating birds and mammals, due to the bioaccumulative* processes. Flame retardants have been found in human blood and researchers fear that disturbances in foetus development may occur.

TCO'99 demand requires that plastic components weighing more than 25 grams must not contain flame retardants with organically bound chlorine and bromine. Flame retardants are allowed in the printed circuit boards since no substitutes are available.

Lead**

Lead can be found in picture tubes, display screens, solders and capacitors. Lead damages the nervous system and in higher doses, causes lead poisoning.

TCO'99 requirement permits the inclusion of lead since no replacement has yet been developed.

Cadmium**

Cadmium is present in rechargeable batteries and in the colour-generating layers of certain computer displays. Cadmium damages the nervous system and is toxic in high doses.

TCO'99 requirement states that batteries, the colour-generating layers of display screens and the electrical or electronics components must not contain any cadmium.

Mercury**

Mercury is sometimes found in batteries, relays and switches. Mercury damages the nervous system and is toxic in high doses.

TCO'99 requirement states that batteries may not contain any Mercury. It also demands that no mercury is present in any of the electrical or electronics components associated with the display unit.

CFCs (freons)

CFCs (freons) are sometimes used for washing printed circuit boards. CFCs break down ozone and thereby damage the ozone layer in the stratosphere, causing increased reception on Earth of ultraviolet light with consequent increased risks of skin cancer (malignant melanoma).

The relevant TCO'99 requirement; Neither CFCs nor HCFCs may be used during the manufacturing and assembly of the product or its packaging.

*Bio-accumulative is defined as substances which accumulate within living organisms.

**Lead, Cadmium and Mercury are heavy metals which are Bio-accumulative.

To obtain complete information on the environmental criteria document, order from:

TCO Development Unit

SE-114 94 Stockholm

SWEDEN

FAX Number: +46 8 782 92 07

E-mail (Internet): development@tco.se

You may also obtain current information on TCO'99 approved and labelled products by visiting their website at:
<http://www.tco-info.com/>

Index

Warning	6
Quick Start	8
Controls.....	11
Recommended Use.....	14
Troubleshooting	15
Specifications	16
Features	17



WARNING



TO PREVENT FIRE OR SHOCK HAZARDS, DO NOT EXPOSE THIS UNIT TO RAIN OR MOISTURE. ALSO, DO NOT USE THIS UNIT'S POLARIZED PLUG WITH AN EXTENSION CORD RECEPTACLE OR OTHER OUTLETS UNLESS THE PRONGS CAN BE FULLY INSERTED.
REFRAIN FROM OPENING THE CABINET AS THERE ARE HIGH VOLTAGE COMPONENTS INSIDE. REFER SERVICING TO QUALIFIED SERVICE PERSONNEL.



CAUTION



CAUTION: TO REDUCE THE RISK OF ELECTRIC SHOCK, MAKE SURE POWER CORD IS UNPLUGGED FROM WALL SOCKET. TO FULLY DISENGAGE THE POWER TO THE UNIT, PLEASE DISCONNECT THE POWER CORD FROM THE AC OUTLET. DO NOT REMOVE COVER (OR BACK). NO USER SERVICEABLE PARTS INSIDE. REFER SERVICING TO QUALIFIED SERVICE PERSONNEL.



This symbol warns user that uninsulated voltage within the unit may have sufficient magnitude to cause electric shock. Therefore, it is dangerous to make any kind of contact with any part inside this unit.



This symbol alerts the user that important literature concerning the operation and maintenance of this unit has been included. Therefore, it should be read carefully in order to avoid any problems.

Declaration of the Manufacturer

We hereby certify that the color monitor
MultiSync® LCD1880SX™ is in
compliance with
Council Directive 73/23/EEC:
– EN 60950

Council Directive 89/336/EEC:
– EN 55022
– EN 61000-3-2
– EN 61000-3-3
– EN 55024

and marked with



NEC-Mitsubishi Electric Visual
Systems Corporation
4-13-23, Shibaura,
Minato-Ku
Tokyo 108-0023, Japan



Power consumption of the monitor is less
than 2 W when in power-saving mode.

Canadian Department of Communications Compliance Statement

DOC: This Class B digital apparatus meets all requirements of the Canadian Interference-Causing Equipment Regulations.

C-UL: Bears the C-UL Mark and is in compliance with Canadian Safety Regulations according to CAN/CSA C22.2 No. 950.

FCC Information

1. Use the attached specified cables with the MultiSync LCD1880SX color monitor so as not to interfere with radio and television reception.
 - (1) Please use the supplied power cord or equivalent to ensure FCC compliance.
 - (2) Please use the specified shielded video signal cable, 15-pin mini D-SUB to 15-pin mini D-SUB cable, 15-pin mini D-SUB to DVI-I cable or DVI-D to DVI-D cable.
2. This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses, and can radiate radio frequency energy, and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:
 - Reorient or relocate the receiving antenna.
 - Increase the separation between the equipment and receiver.
 - Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
 - Consult your dealer or an experienced radio/TV technician for help.

If necessary, the user should contact the dealer or an experienced radio/television technician for additional suggestions. The user may find the following booklet, prepared by the Federal Communications Commission, helpful: "How to Identify and Resolve Radio-TV Interference Problems." This booklet is available from the U.S. Government Printing Office, Washington, D.C., 20402, Stock No. 004-000-00345-4.

DECLARATION OF CONFORMITY

This device complies with Part 15 of FCC Rules. Operation is subject to the following two conditions. (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

U.S. Responsible Party:	NEC-Mitsubishi Electronics Display of America, Inc.
Address:	1250 North Arlington Heights Road, Suite 500
	Itasca, Illinois 60143-1248
Tel. No.:	(630) 467-3000

Type of Product:	Display Monitor
Equipment Classification:	Class B Peripheral
Model:	MultiSync LCD1880SX



*We hereby declare that the equipment specified above
conforms to the technical standards as specified in the FCC Rules.*

Windows is a registered trademark of Microsoft Corporation. NEC is a registered trademark of NEC Corporation. ENERGY STAR is a U.S. registered trademark. All other brands and product names are trademarks or registered trademarks of their respective owners.

As an ENERGY STAR® Partner, NEC-Mitsubishi Electronics Display of America has determined that this product meets the Energy Star guidelines for energy efficiency. The ENERGY STAR emblem does not represent EPA endorsement of any product or service.

Quick Start

To attach the MultiSync® LCD monitor to your system, follow these instructions:

1. Turn off the power to your computer.
2. For the PC or MAC with DVI digital output: Connect the DVI signal cable to the connector of the display card in your system (**Figure A.1**). Tighten all screws.
For the PC with Analog output: Connect the 15-pin mini D-SUB to DVI-A signal cable to the connector of the display card in your system (**Figure A.2**).
For the MAC: Connect the MultiSync Macintosh cable adapter to the computer, then attach the 15-pin mini D-SUB signal cable to the MultiSync Macintosh cable adapter (**Figure B.1**).

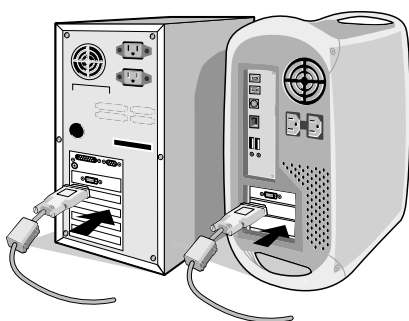


Figure A.1

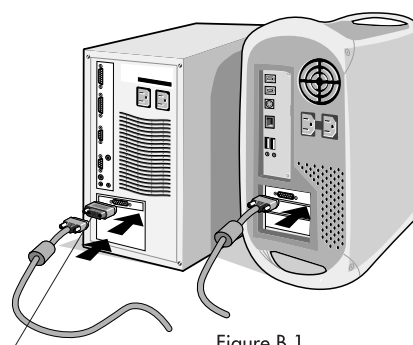


Figure B.1

Macintosh
Adapter
(not included)

Macintosh Cable Adapter (not included)
Note: Some Macintosh systems do not
require a Macintosh Cable Adapter

NOTE: Some Macintosh systems do not require a Macintosh cable adapter.

3. Connect the DVI signal cable to the connector on the back of the monitor. Place the video signal cable (**Figure C.1**). Connect only either VGA or DVI to input 2.

NOTE: Incorrect cable connections may result in irregular operation, damage display quality/components of LCD module and/or shorten the module's life.

Collect cables and keep them in the stand with attached wirecramps. Use 2 wirecramps for keep them in the back stand (**Figure C.1**), or use 1 wirecramp for keep them in the front stand (**Figure C.2**).

Please check Tilt, Rise and Lower monitor screen and screen rotation when you manage cables.

4. Connect one end of the power cord to the AC inlet on the back of the monitor and the other end to the power outlet (**Figure C.1**).

NOTE: Please use the supplied AC Power Cord unless you use this monitor in foreign country (including the user in U.S.A. or Canada uses this monitor in AC 126-240V power supply). Please refer to Recommended Use section for proper selection of AC Power cord.

5. The Vacation Switch on the left side of the monitor must be turned on. Turn on the monitor with the front power button (**Figure D.1**) and the computer.

NOTE: There are two switches: one on the left side and on the front side of the monitor. DO NOT switch on/off quickly.

6. To complete the setup of your MultiSync LCD monitor, use the following OSM™ controls:

- Auto Adjust Contrast (Analog input only)
- Auto Adjust (Analog input only)

Refer to the Controls section of this User's Manual for a full description of these OSM controls.

NOTE: If you have any problems, please refer to the **Troubleshooting** section of this User's Manual.

NOTE: Refer to User's Manual in the NEC LCD Setup Software CD case for installation and operation of this software.

Quick Start –continued

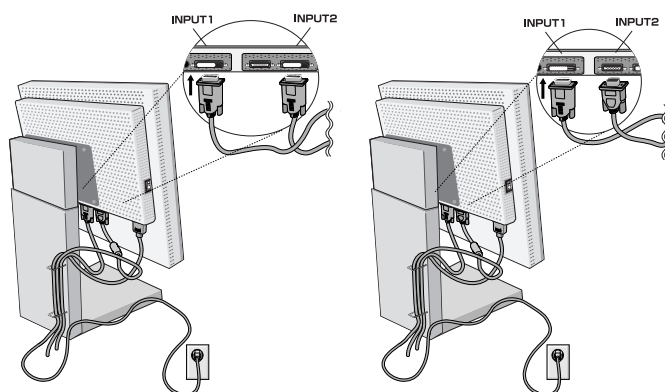
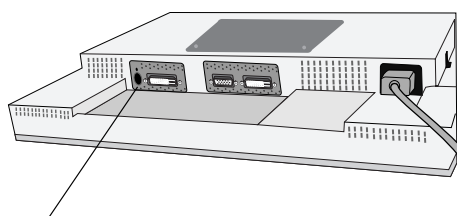


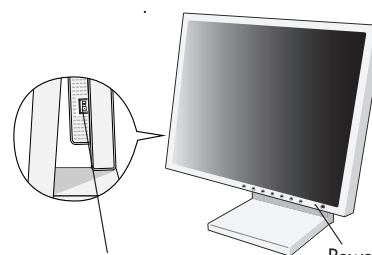
Figure C.1



Figure C.2



This connector is for the specified option.
Don't connect a product which is not specified.



Vacation
Switch

Power
Button

Figure D.1

Raise and Lower Monitor Screen

The monitor may be raised or lowered in either Portrait or Landscape mode.
To raise or lower screen, place hands on each side of the monitor and lift or lower to the desired height (Figure RL.1).

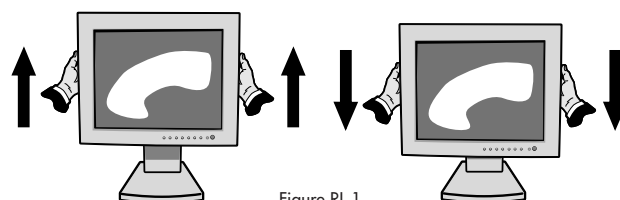


Figure RL.1

Screen Rotation

Before rotating, the screen must be raised to the highest level to avoid knocking the screen on the desk or pinching your fingers.

To raise the screen, place hands on each side of the monitor and lift up to the highest position (Figure RL.1).

To rotate screen, place hands on each side of the monitor screen and turn clockwise from Landscape to Portrait or counter-clockwise from Portrait to Landscape (Figure R.1).

To toggle the orientation of the OSM menu between Landscape and Portrait modes, press the RESET button while OSM menu is off.

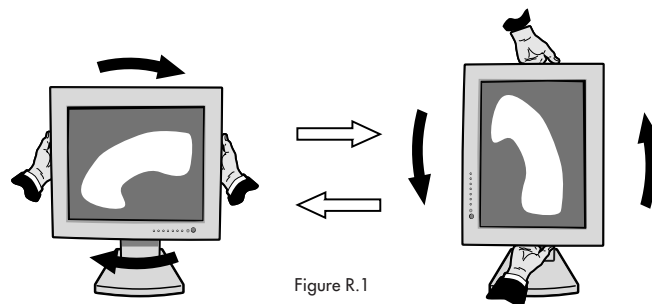


Figure R.1

Quick Start –continued

Tilt

Grasp top and bottom sides of the monitor screen with your hands and adjust the tilt as desired (**Figure TS.1**).



Figure TS.1

Swivel

Grasp both sides of the monitor screen with your hands and adjust the swivel as desired (**Figure TS.2**).



Figure TS.2

Remove Monitor Stand for Mounting

To prepare the monitor for alternate mounting purposes:

1. Disconnect all cables.
2. Place hands on each side of the monitor and lift up to the highest position .
3. Place monitor face down on a non-abrasive surface. (Place the screen on a 45 mm / 1.8 inch platform so that the stand is parallel with the surface.) (**Figure S.1**)

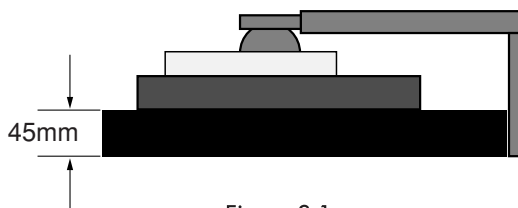


Figure S.1

4. Remove the 4 screws connecting the monitor to the stand and lift off the stand assembly (**Figure S.3**). The monitor is now ready for mounting in an alternate manner.
5. Reverse this process to reattach stand.

NOTE: Use only VESA-compatible alternative mounting method.(100mm pitch)

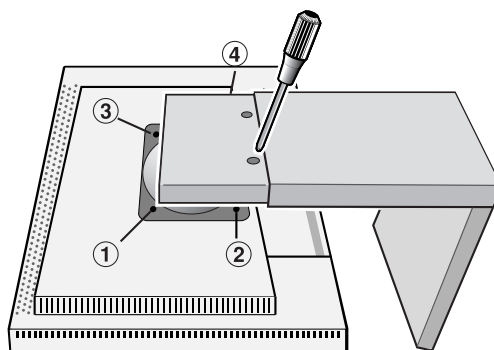


Figure S.2

Caution: Please use the attached screws (4 pcs) when mounting. To fulfil the safety requirements the monitor must be mounted to an arm which guaranties the necessary stability under consideration of the weight of the monitor. The LCD monitor shall only be used with an approved arm (e.g. GS mark).

Controls

OSM™ (On-Screen Manager) control buttons on the front of the monitor function as follows:

To access OSM menu, press any of the control buttons (◀, ▶, -, +).

To change signal input, press the SELECT button.

NOTE: OSM must be closed in order to change signal input.

	Menu
EXIT	Exits the OSM controls. Exits to the OSM main menu.
CONTROL ◀ / ▶	Moves the highlighted area left/right to select control menus. Moves the highlighted area up/down to select one of the controls.
ADJUST -/+	Moves the bar left/right to increase or decrease the adjustment. Active Auto Adjust function. Enter the Sub Menu.
SELECT	Enter the OSM controls. Enter the OSM sub menu.
RESET	Resets the highlighted control menu to the factory setting.

NOTE: When **RESET** is pressed in the main and sub-menu, a warning window will appear allowing you to cancel the **RESET** function by pressing the EXIT button.

☀️ **Brightness/Contrast Controls**

☀️ **BRIGHTNESS**

Adjusts the overall image and background screen brightness.

🔵 **CONTRAST**

Adjusts the image brightness in relation to the background.

AUTO **AUTO ADJUST CONTRAST**

Adjusts the image displayed for non-standard video inputs.

☐ **AUTO BRIGHTNESS**

This function is to adjust the brightness automatically to the extent of white display area, against dazzle.

AUTO **Auto Adjust**

Automatically adjusts the Image Position and H. Size settings and Fine settings.

☐ **Position Controls**

☐ **LEFT / RIGHT**

Controls Horizontal Image Position within the display area of the LCD.

☐ **DOWN / UP**

Controls Vertical Image Position within the display area of the LCD.

☐ **H.SIZE**

Adjusts the horizontal size by increasing or decreasing this setting.

⌘ **FINE**

Improves focus, clarity and image stability by increasing or decreasing this setting.

☐ **AUTO FINE**

This function is to adjust "FINE" automatically and periodically (once per 32 min.) against changes of the signal condition or ambient temperature and to adjust "FINE" automatically when changing the signal timing.

RGB AccuColor® Control Systems (For North America Only)

AccuColor® Control Systems: Six color presets select the desired color setting. Color temperature increases or decreases in each preset. (NATIVE color preset cannot be changed.)

NATIVE: Original color presented by the LCD panel that is unadjustable.

Controls –continued

Tools 1

 **SHARPNESS:** This function is digitally capable to keep crisp image at any timings. It is continuously adjustable to get distinct image or soft one as you prefer, and set independently by different timings. The number of adjustment steps is different at the case that input is a digital or analog.

 **EXPANSION MODE:** Sets the zoom method.

FULL: The image is expanded to 1280 x 1024, regardless of the resolution.

ASPECT: The image is expanded without changing the aspect ratio.

OFF: The image is not expanded.

CUSTOM (Digital input and Resolution of 1280 x 1024 only):

Select one of seven expansion rates. In this mode the resolution may be low and there may be blank areas. This mode is for use with special video cards.

 **VIDEO DETECT:** Selects the method of video detection when more than one computer is connected.

FIRST DETECT: The video input has to be switched to "FIRST DETECT" mode. When current video input signal is not present, then the monitor searches for a video signal from the other video input port. If the video signal is present in the other port, then the monitor switches the video source input port to the new found video source automatically. The monitor will not look for other video signals while the current video source is present.

LAST DETECT: The video input has to be switched to the "LAST DETECT" mode. When the monitor is displaying a signal from the current source and a new secondary source is supplied to the monitor, then the monitor will automatically switch to the new video source. When current video input signal is not present, then the monitor searches for a video signal from the other video input port. If the video signal is present in the other port, then the monitor switches the video source input port to the new found video source automatically.

NONE: The Monitor will not search the other video input port unless the monitor is turned on.

 **DVI SELECTION:** This function selects the DVI input mode (Input 1). When the DVI selection has been change, you must restart your computer.

☐ **OFF TIMER:** Monitor will automatically Turned off after a period of setting time from power on with ON/OFF function.

Tools 2

 **LANGUAGE:** OSM™ control menus are available in seven languages.

 **OSM POSITION:** You can choose where you would like the OSM control image to appear on your screen. Selecting OSM Location allows you to manually adjust the position of the OSM control menu left, right, down or up.

 **OSM TURN OFF:** The OSM control menu will stay on as long as it is use. In the OSM Turn Off submenu, you can select how long the monitor waits after the last touch of a button to shut off the OSM control menu. The preset choices are 10, 20, 30, 45, 60 and 120 seconds.

 **OSM LOCK OUT:** This control completely locks out access to all OSM control functions. When attempting to activate OSM controls while in the Lock Out mode, a screen will appear indicating the OSM controls are locked out. To activate the OSM Lock Out function, press ◀, then ▶ and hold down simultaneously. To de-activate the OSM Lock Out, press ◀, then ▶ and hold down simultaneously.

☐ **OSM ROTATE:** To rotate OSM between Landscape and Portrait modes.

 **RESOLUTION NOTIFIER:** This optimal resolution is 1280 x 1024. If ON is selected, a message will appear on the screen after 30 seconds, notifying you that the resolution is not at 1280 x 1024.

☐ **HOT KEY:** You can adjust the brightness and contrast directly. When this function is set to ON, you can adjust the brightness with ◀ or ▶, contrast with + or - key, while the OSM menu is off.

 **FACTORY PRESET:** Selecting Factory Preset allows you to reset all OSM control settings back to the factory settings. Individual settings can be reset by highlighting the control to be reset and pressing the RESET button.

Controls –continued

Information

MODE DISPLAY MODE: Provides information about the current resolution display and technical data including the preset timing being used and the horizontal and vertical frequencies. Increases or decreases the current resolution.



MONITOR INFO: Indicates the model and serial numbers of your monitor.

OSM™ Warning: OSM Warning menus disappear with Exit button.

NO SIGNAL: This function gives a warning when there is no Horizontal or Vertical Sync. After power is turned on or when there is a change of input signal, the **No Signal** window will appear.

RESOLUTION NOTIFIER: This function gives a warning of use with optimized resolution. After power is turned on or when there is a change of input signal or the video signal doesn't have proper resolution, the **Resolution Notifier** window will open. This function can be disabled in the TOOL menu.

OUT OF RANGE: This function gives a recommendation of the optimized resolution and refresh rate. After the power is turned on or there is a change of input signal or the video signal doesn't have proper timing, the **Out Of Range** menu will appear.

CHECK CABLE: This function will advise you to check all Video Inputs on the monitor and computer to make sure they are properly connected.

NOTE: If “ CHANGE DVI SELECTION” is displayed, switch to DVI SELECTION.



Recommended Use

Safety Precautions and Maintenance



FOR OPTIMUM PERFORMANCE, PLEASE NOTE THE FOLLOWING WHEN SETTING UP AND USING THE MULTISYNC® LCD COLOR MONITOR:



- **DO NOT OPEN THE MONITOR.** There are no user serviceable parts inside and opening or removing covers may expose you to dangerous shock hazards or other risks. Refer all servicing to qualified service personnel.
- Do not spill any liquids into the cabinet or use your monitor near water.
- Do not insert objects of any kind into the cabinet slots, as they may touch dangerous voltage points, which can be harmful or fatal or may cause electric shock, fire or equipment failure.
- Do not place any heavy objects on the power cord. Damage to the cord may cause shock or fire.
- Do not place this product on a sloping or unstable cart, stand or table, as the monitor may fall, causing serious damage to the monitor.
- The power supply cord you use must have been approved by and comply with the safety standards of your country. (Type H05VV-F should be used in Europe), and matches the power supply voltage of the AC outlet being used.
- In UK, use a BS-approved power cord with molded plug having a black (5A) fuse installed for use with this monitor. If a power cord is not supplied with this monitor, please contact your supplier.
- Do not place any objects onto the monitor and do not use the monitor outdoors.
- The inside of the fluorescent tube located within the LCD monitor contains mercury. Please follow the bylaws or rules of your municipality to dispose of the tube properly.

Immediately unplug your monitor from the wall outlet and refer servicing to qualified service personnel under the following conditions:

- When the power supply cord or plug is damaged.
- If liquid has been spilled, or objects have fallen into the monitor.
- If the monitor has been exposed to rain or water.
- If the monitor has been dropped or the cabinet damaged.
- If the monitor does not operate normally by following operating instructions.
- Do not bend power cord.
- Do not use monitor in high temperature, humid, dusty, or oily areas.
- If glass is broken, handle with care.
- Do not cover vent on monitor.
- If monitor or glass is broken, do not come in contact with the liquid crystal and handle with care.



CAUTION

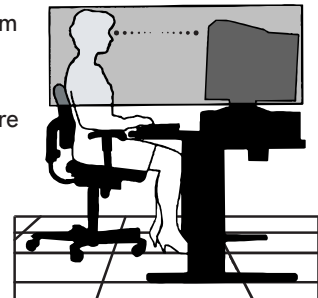
- Allow adequate ventilation around the monitor so that heat can properly dissipate. Do not block ventilated openings or place the monitor near a radiator or other heat sources. Do not put anything on top of monitor.
- The power cable connector is the primary means of detaching the system from the power supply. The monitor should be installed close to a power outlet which is easily accessible.
- Handle with care when transporting. Save packaging for transporting.



CORRECT PLACEMENT AND ADJUSTMENT OF THE MONITOR CAN REDUCE EYE, SHOULDER AND NECK FATIGUE. CHECK THE FOLLOWING WHEN YOU POSITION THE MONITOR:



- For optimum performance, allow 20 minutes for warm-up.
- Adjust the monitor height so that the top of the screen is at or slightly below eye level. Your eyes should look slightly downward when viewing the middle of the screen.
- Position your monitor no closer than 16 inches and no further away than 28 inches from your eyes. The optimal distance is 23 inches.
- Rest your eyes periodically by focusing on an object at least 20 feet away. Blink often.
- Position the monitor at a 90° angle to windows and other light sources to minimize glare and reflections. Adjust the monitor tilt so that ceiling lights do not reflect on your screen.
- If reflected light makes it hard for you to see your screen, use an anti-glare filter.
- Clean the LCD monitor surface with a lint-free, non-abrasive cloth. Avoid using any cleaning solution or glass cleaner!
- Adjust the monitor's brightness and contrast controls to enhance readability.
- Use a document holder placed close to the screen.
- Position whatever you are looking at most of the time (the screen or reference material) directly in front of you to minimize turning your head while you are typing.
- Avoid displaying fixed patterns on the monitor for long periods of time to avoid image persistence (after-image effects).





Recommended Use –continued

- Get regular eye checkups.

Ergonomics

To realize the maximum ergonomics benefits, we recommend the following:

- Adjust the Brightness until the background raster disappears
- Do not position the Contrast control to its maximum setting
- Use the preset Size and Position controls with standard signals
- Use the preset Color Setting
- Use non-interlaced signals with a vertical refresh rate between 60-75Hz
- Do not use primary color blue on a dark background, as it is difficult to see and may produce eye fatigue to insufficient contrast

Troubleshooting

No picture

- The signal cable should be completely connected to the display card/computer.
- The display card should be completely seated in its slot.
- Check the Vacation Switch should be in the ON position.
- Front Power Switch and computer power switch should be in the ON position.
- Check to make sure that a supported mode has been selected on the display card or system being used. (Please consult display card or system manual to change graphics mode.)
- Check the monitor and your display card with respect to compatibility and recommended settings.
- Check the signal cable connector for bent or pushed-in pins.
- Check the signal input, "INPUT 1" or "INPUT 2".

Power Button does not respond

- Unplug the power cord of the monitor from the AC outlet to turn off and reset the monitor, or simultaneously press the RESET and Power buttons.
- Check the Vacation Switch on the right side of the monitor.

Image persistence

- Image persistence is when a "ghost" of an image remains on the screen even after the monitor has been turned off. Unlike CRT monitors, LCD monitors' image persistence is not permanent. To alleviate image persistence, turn the monitor off for as long as an image was displayed. If an image was on the monitor for one hour and a "ghost" of that image remains, the monitor should be turned off for one hour to erase the image.

NOTE: As with all personal display devices, NEC-Mitsubishi Electronics Display recommends using a screen saver at regular intervals whenever the screen is idle.

Image is unstable, unfocused or swimming is apparent

- Signal cable should be completely attached to the computer.
- Use the OSM Image Adjust controls to focus and adjust display by increasing or decreasing the fine total. When the display mode is changed, the OSM Image Adjust settings may need to be re-adjusted.
- Check the monitor and your display card with respect to compatibility and recommended signal timings.
- If your text is garbled, change the video mode to non-interlace and use 60Hz refresh rate.

LED on monitor is not lit (no green or amber color can be seen)

- Power Switch should be in the ON position and power cord should be connected.

Display image is not sized properly

- Use the OSM Image Adjust controls to increase or decrease the Coarse total.
- Check to make sure that a supported mode has been selected on the display card or system being used. (Please consult display card or system manual to change graphics mode.)

No Video

- If no video is present on the screen, turn the Power button off and on again.
- Make certain the computer is not in a power-saving mode (touch the keyboard or mouse).

If you have a problem, Please contact to us.

Specifications

Monitor Specifications		MultiSync® LCD1880SX™	Notes
LCD Module	Diagonal: Viewable Image Size: Native Resolution (Pixel Count):	18.1 inch 18.1 inch Landscape: 1280x1024 Portrait: 1024 x 1280	Active matrix; thin film transistor (TFT) liquid crystal display (LCD); 0.28 mm dot pitch; 240cd/m² white luminance; 300:1 contrast ratio, typical
Input Signal	Video: Sync:	ANALOG 0.7 Vp-p/75 Ohms Separate sync. TTL Level Horizontal sync. Positive/Negative Vertical sync. Positive/Negative Composite sync. Positive/Negative Sync on Green (Video 0.7p-p and Sync Negative 0.3V p-p)	Digital Input: TMDS™
Display Colors	Analog input:	Over 16 million	Depends on display card used.
Synchronization Range	Horizontal: Vertical:	31.0 kHz to 82 kHz 50 Hz to 85 Hz	Automatically Automatically
Image Formation Time		50 ms (Typ.) 59 ms (Max.)	
Resolutions Supported	Landscape: Portrait:	720 x 400*¹ at 70 Hz to 85 Hz 640 x 480*¹ at 60 Hz to 85 Hz 800 x 600*¹ at 56 Hz to 85 Hz 832 x 624*¹ at 75 Hz 1024 x 768*¹ at 60 Hz to 85 Hz 1152 x 870*¹ at 75Hz 1280 x 1024 at 60 Hz to 75 Hz 480 x 640*¹ at 60 Hz to 85 Hz 600 x 800*¹ at 56 Hz to 85 Hz 624 x 832*¹ at 75 Hz 768 x 1024*¹ at 60 Hz to 85 Hz 870 x 1152*¹ at 75Hz 1024 x 1280 at 60 Hz to 75 Hz	Some systems may not support all modes listed. NEC-Mitsubishi Electronics Display cites recommended resolution at 60 Hz for optimal display performance. Some systems may not support all modes listed. NEC-Mitsubishi Electronics Display cites recommended resolution at 60 Hz for optimal display performance.
Active Display Area	Landscape : Horiz. : Vert. : Horizontal : Horiz. : Vert. :	359 mm/14.1 inches 287 mm/11.3 inches 287 mm/11.3 inches 359 mm/14.1 inches	
Power Supply		AC 100 - 240 V @ 50/60 Hz	
Current Rating		0.65 A @ 100-0.30 A @ 240 V	
Dimensions	Landscape: Portrait: Without stand: Height Adjustment:	387.0 mm (W) x 380~490 mm (H) x 180.0 mm (D) 15.2 inches (W) x 15.0~19.3 inches (H) x 7.1 inches (D) 315.2mm (W) x 416~526 mm (H) x 180.0 mm (D) 12.4 inches (W) x 16.4~20.7inches (H) x 7.1 inches (D) 387mm (W) x 315.2 mm (H) x 61.7 mm (D) 15.2 inches (W) x 12.4inches (H) x 2.4 inches (D) 110mm	
Weight	Without stand:	8.5 kg 18.7 lbs 5.5 kg 12.1 lbs	
Environmental Considerations	Operating Temperature: Humidity: Feet: Storage Temperature: Humidity: Feet:	5°C to 35°C/41°F to 95°F 30% to 80% 0 to 10,000 Feet -10°C to 60°C/14°F to 140°F 10% to 85% 0 to 45,000 Feet	

*¹ Interpolated Resolutions: When resolutions are shown that are lower than the pixel count of the LCD module, text may appear different. This is normal and necessary for all current flat panel technologies when displaying non-native resolutions full screen. In flat panel technologies, each dot on the screen is actually one pixel, so to expand resolutions to full screen, an interpolation of the resolution must be done.
NOTE: Technical specifications are subject to change without notice.



Features

ambix™ Technology: Dual input technology allowing both analog and digital inputs off of one connector (DVI-I) as well as additional legacy analog support off of a traditional 15-pin VGA connector. Provides traditional MultiSync® technology compatibility for analog as well as TMDS™-based (Transition Minimized Differential Signaling) digital compatibility for digital inputs. TMDS-based digital interfaces include DVI-D, DFP and P&D.

DVI-I: The integrated interface ratified by the Digital Display Working Group (DDWG) that allows both digital and analog connectors off of one port. The "I" stands for integration for both digital and analog. The digital portion is TMDS-based.

DVI-D: The digital-only subset of DVI ratified by the Digital Display Working Group (DDWG) for digital connections between computers and displays. As a digital-only connector, analog support is not provided off a DVI-D connector. As a TMDS-based digital only connection, only a simple adapter is necessary for compatibility between DVI-D and other TMDS-based digital connectors such as DFP and P&D.

DFP (Digital Flat Panel): An all-digital interface for flat panel monitors which is signal compatible with DVI. As a TMDS-based digital only connection, only a simple adapter is necessary for compatibility between DFP and other TMDS-based digital connectors such as DVI and P&D.

P&D (Plug and Display): The VESA standard for digital flat panel monitor interfaces. It is more robust than DFP since it allows for other options off a signal connector (options like USB, analog video and IEEE-1394-995). The VESA committee has recognized that DFP is a subset of P&D. As a TMDS-based connector (for the digital input pins), only a simple adapter is necessary for compatibility between P&D and other TMDS-based digital connector such as DVI and DFP.

Pivoting Stand: Allows users to adjust the monitor to the orientation that best fits their application, either Landscape orientation for wide documents, or portrait orientation for the ability to preview a full page on one screen at one time. The Portrait orientation is also perfect for full screen video conferencing.

Reduced Footprint: Provides the ideal solution for environments requiring superior image quality but with size and weight limitations. The monitor's small footprint and low weight allow it to be moved or transported easily from one location to another.

AccuColor® Control Systems (For North America Only): Six color presets select the desired color setting (sRGB and NATIVE color presets are standard and cannot be changed). Color temperature increases or decreases, in each preset. R,Y,G,C,B,M,S: Increases or decreases Red, Yellow, Green, Cyan, Blue, Magenta and Saturation depending upon which is selected. The change in color will appear on screen and the direction (increase or decrease) will be shown by the color bars. NATIVE: Original color presented by the LCD panel that is unadjustable.

OSM™ (On-Screen Manager) Controls: Allow you to quickly and easily adjust all elements of your screen image via simple to use on-screen menus.

ErgoDesign® Features: Enhance human ergonomics to improve the working environment, protect the health of the user and save money. Examples include OSM controls for quick and easy image adjustments, tilt base for preferred angle of vision, small footprint and compliance with MPRII and TCO guidelines for lower emissions.

Plug and Play: The Microsoft® solution with the Windows®95/98/2000 operating system facilitates setup and installation by allowing the monitor to send its capabilities (such as screen size and resolutions supported) directly to your computer, automatically optimizing display performance.

IPM™ (Intelligent Power Manager) System: Provides innovative power-saving methods that allow the monitor to shift to a lower power consumption level when on but not in use, saving two-thirds of your monitor energy costs, reducing emissions and lowering the air conditioning costs of the workplace.

Multiple Frequency Technology: Automatically adjusts monitor to the display card's scanning frequency, thus displaying the resolution required.

FullScan™ Capability: Allows you to use the entire screen area in most resolutions, significantly expanding image size.

XtraView+® Wide Viewing Angle Technology: Allows the user to be able to see the monitor from any angle (170 degrees) from any orientation — Portrait or Landscape. Provides full 170° viewing angles either up, down, left or right.

VESA Standard Mounting Interface: Allows users to connect their MultiSync monitor to any VESA standard third party mounting arm or bracket. Allows for the monitor to be mounted on a wall or an arm using any third party compliant device.

OSM Display Screen Copyright 2002 by NEC-Mitsubishi Electronics Display of America, Inc.